



Project Proposal

Charity Operations Management Database

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Selected problem and context

Charity organizations typically run multiple **campaigns** (e.g., seasonal fundraising, food support, winter kits) and execute **events** (community engagement, distributions, volunteer trainings). They receive donations from **donors** (money and/or in-kind), collaborate with **partners** (restaurants, sponsors, suppliers), and serve **beneficiaries** through aid requests and distributions. In many small-to-mid regional teams, operational data is spread across spreadsheets, online forms, email threads, and chat messages. This fragmentation creates recurring operational problems, including duplicate or inconsistent donor/beneficiary records, weak traceability from donations to campaigns/events and onward to distributions and beneficiaries, scheduling conflicts caused by inconsistent shift assignments and unreliable attendance tracking, and reporting gaps that make it slow and error-prone to produce accurate totals for funds raised, partner contributions, volunteer hours, and beneficiaries served.

Project goal: design and implement a relational database that centralizes campaign/event operations, donation tracking, partner contributions, beneficiary servicing, and volunteer/staff scheduling enabling reliable integrity constraints and standard reporting queries.

This problem is inspired by a **real-world** charity operations scenario (PEI Islamic Relief regional charity team workflows), but it is described at an abstract level and does not include any real names, communications, or operationally sensitive details.

Why a relational database is appropriate

A relational database is a strong fit because charity operations naturally involve multiple entities and relationships (donors–donations, campaigns–events, events–shifts, volunteers–shifts, beneficiaries–requests–distributions, partners–contributions). These relationships require:

- **Primary keys and foreign keys** to prevent orphan records (e.g., a donation must belong to a valid donor)
- **Many-to-many modeling** via junction tables (e.g., volunteers assigned to many shifts; shifts include many volunteers)
- **Constraints** to enforce business rules (e.g., donation amounts must be positive; event shift capacity cannot be exceeded)
- **Join-based reporting** for operational and accountability needs (e.g., totals raised per campaign, beneficiaries served per event, volunteer hours by month)

In short, the database must enforce structure and integrity while supporting complex queries across linked data exactly what relational modeling is designed for.

Data representation using synthetic examples

No real or sensitive data will be used or submitted. All records will be synthetic and generated solely for the purpose of the course project. For example:

- Donors will be represented with fictional names or donor codes (e.g., DNR-10021) and placeholder contact details.

- Beneficiaries will be represented using a non-identifying synthetic code (e.g., BEN-00418), broad area/region (e.g., “Charlottetown Area”), and non-sensitive household attributes (e.g., household size).
- Partners will be fictional organizations (e.g., “Northside Catering”) with synthetic contacts.
- Donations and distributions will be generated to reflect realistic volumes and patterns (e.g., spikes during major campaigns/events), while remaining entirely synthetic.

Feasibility within the course timeline

This project is feasible within the course scope because:

- The domain is well-bounded and can be implemented incrementally (campaigns/events → donations → beneficiaries → scheduling).
- The ERD and schema are straightforward to normalize (clear junction tables for M:N relationships).
- Synthetic data generation is manageable (dozens of campaigns/events, hundreds of donations, and a scheduling dataset for volunteers/staff).

Planned implementation approach:

1. ERD + relational schema (PK/FK, constraints, normalization)
2. Populate synthetic dataset and validate constraints
3. Implement a set of core SQL queries/reports (campaign totals, event performance, volunteer hours, partner contributions, beneficiary service counts)
4. Refinement and final documentation

This aligns with the requirement to be feasible “within the scope of this course” and timeline.

How the problem was identified and AI usage

Our team has direct experience working with charity organizations and understands the operational workflow involved in managing donors and donations, running campaigns and events, coordinating partners, serving beneficiaries, and scheduling volunteers/staff. Based on this experience, we identified common operational pain points such as fragmented records across spreadsheets and messages, inconsistent donor/beneficiary tracking, weak traceability from donations to campaign/event outcomes, and difficulty producing accurate reports.

Use of AI tools: AI tools were used as a support resource during brainstorming to propose alternative charity workflow variations, sanity-check that the chosen scope contains sufficient entities and relationships and suggest typical integrity constraints and reporting queries. Final scope decisions, entity selection, and modeling choices were made by the group.